

Shared care notes that make trust visible.

A clinic-scoped longitudinal collaboration layer for the moment when a care team needs to understand what changed and what needs action in under ten seconds.

Phase 4 local synthetic prototype · FastAPI + React/TypeScript · SQLite local target · PostgreSQL path retained but not provisioned

46

real backend tests

97%

backend coverage

78.477 ms

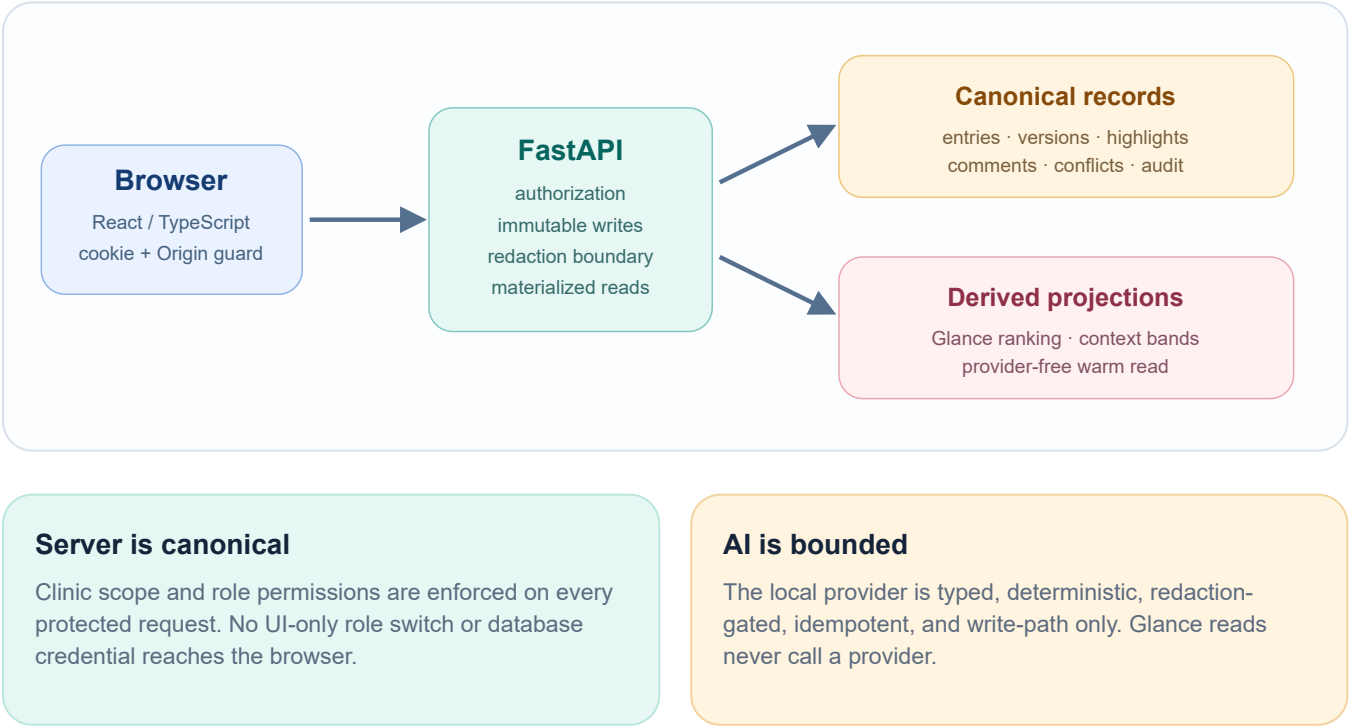
Gate C warm-path P95

SYSTEM BOUNDARY

Product thesis

Nightingale is not an EHR replacement, diagnostic system, or autonomous medical decision-maker. AI-scribed content remains a suggestion until clinician review. Display ranking, explicit medical risk, provenance, and human confirmation are separate fields.

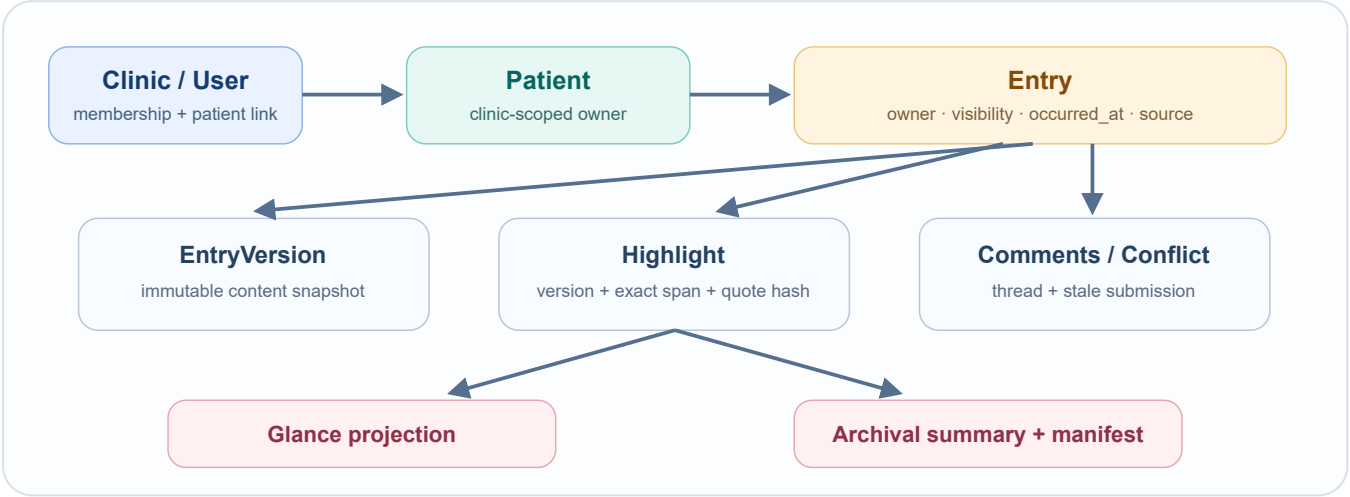
ARCHITECTURE



Canonical evidence stays intact while views stay glanceable.

The design separates stable record identity, immutable snapshots, source-linked highlights, human review, and derivative read models. Summaries are never the source of truth.

SCHEMA RELATIONSHIPS



EXACT PROVENANCE

Source navigation contract

`source\_entry\_id` + `source\_version\_id` + Python Unicode-codepoint offsets + exact quote + SHA-256. The frontend validates `Array.from(text)` and shows an integrity warning instead of an approximate match.

TRUST CONTROLS

Human decisions remain explicit

Suggested, accepted, rejected, superseded, and conflict-review states remain visible. A clinician can confirm semantic truth, but cannot erase the source record.

HYBRID RETRIEVAL BANDS

BAND	POLICY	READ SHAPE
Hot	Last 14 days, or protected by action/risk/conflict/discussion/pin/confirmation/care plan.	Full current detail.
Warm	14–90 days and not protected.	Metadata index; canonical source remains available.
Cold	Older than 90 days and not protected.	Derived period summary, manifest hash, and immutable source pointers.

```
final display priority = clamp(base + recency + explicit risk + open action + clinician confirmation + adaptive feedback, 0, 100)
```

Adaptive feedback is clinic-scoped, idempotent, structured-feature-only, and bounded to [-12, +12]. It changes display priority, never risk or provenance.

A reproducible local prototype with an honest production boundary.

Evidence below is synthetic and local. It establishes application behavior, not hosted compliance or PostgreSQL production guarantees.

QUALITY GATES

Backend

46 passed · 97% coverage · Ruff · mypy · pip check.
Alembic `0006\_gate\_d\_archival` passes fresh, legacy repair, downgrade/re-upgrade, and `alembic check` paths.

Frontend

8 Vitest tests · ESLint · Prettier · TypeScript · Vite build.
Historical context and ranking explanations are unit-covered.

Browser

8 Playwright checks at 1440×900 and 390×844. Exact source/deep-link, ranking feedback, revisions/comments, 409 conflict, context source pointer, and patient privacy.

Warm path

P95 78.477 ms from 1,000 real TCP requests, concurrency 10, 26 patients, 208 materialized rows, zero errors. Local SQLite/Uvicorn approximation.

THREE-MINUTE DEMONSTRATION PATH

A

Trace and trust

Clinician opens a Glance item, expands **Why ranked?**, pins/unpins it, follows the exact immutable source, and accepts or rejects a suggestion.

B

Collaborate without erasure

Staff edits a role-owned note, compares versions, reverts as a new version, adds a nested internal comment, and resolves/unresolves the thread.

C

Surface conflict and history

Two writes use one expected version; the stale write returns **409** and remains beside the winner. Historical context then discloses a derived summary and opens a canonical source pointer.

TRADE-OFFS AND DEFERRED WORK

What is implemented: server-side RBAC and clinic scope, immutable revisions, exact provenance, comments, conflict preservation, redaction-gated deterministic AI jobs, bounded adaptive ranking, derived archival context, and materialized warm reads.

What is not claimed: live external LLM quality, PostgreSQL execution, TLS/encryption-at-rest evidence, production retention policy, human 10-second UX sign-off, voice capture, deployment, or email submission.

Safety sentence

AI output is a suggestion. Derived summaries are not canonical sources. Risk is not learned silently. Human review is not replaced by ranking.